A

Mini Project Report on

MEDLY

Submitted in partial fulfillment of the requirements for the degree

Second Year Engineering – Computer Science Engineering (Data Science)

by

Bhumika Masane 24107039

Arju Pandey 24107028

Shreya Pandit 24107001

Shravani Bhosale 24107025

Under the guidance of

Prof. Ashwini Parkar

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING (DATA SCIENCE)
A.P. SHAH INSTITUTE OF TECHNOLOGY
G.B. Road, Kasarvadavali, Thane (W)-400615
UNIVERSITY OF MUMBAI

Academic year: 2025-26

CERTIFICATE

This to certify that the Mini Project report on MEDLY has been submitted by Shravni Bhosale (24107025) ,Shreya Pandit (24107001), Arju Pandey (24107028) and Bhumika Masane (24107039) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

Prof. Ashwini P
Guide

Dr. Pravin Adivarekar
HOD, CSE(Data Science)

Dr. Uttam D. Kolekar
Principal

External Examiner:
1.

Internal Examiner:
1.

Place: A. P. Shah Institute of Technology, Thane
Date:

ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide **Prof.Ashwini Parkar.** Expressing gratitude towards our HoD, **Dr. Pravin Adivarekar**, and the Department of Computer Science Engineering (Data Science) for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our project coordinator **Prof. Richa Singh and Prof Kiran Gaikwad** who gave us her valuable suggestions and ideas when we were in need of them. We would also like to thank our peers for their helpful suggestions.

TABLE OF CONTENTS

1. Introduction 5

   1.1. Purpose 5

   1.2. Problem Statement 5

   1.3. Objectives 6

   1.4. Scope 6

2. Proposed System 9

   2.1. Features and Functionality 9

3. Project Outcomes 14

4. Software Requirements 17

5. Project Design 21

6. Project Scheduling 24

7. Results 28

8. Conclusion 39

9. References 40

Chapter 1 : Introduction

In today's fast-paced lifestyle, maintaining proper health has become a major challenge, especially for individuals who rely on regular medication. Patients often face difficulties in remembering their medication schedules, tracking doses, and ensuring timely intake. This issue is more critical for elderly people, individuals with chronic illnesses, and those managing multiple prescriptions simultaneously.

With the advancement of digital technology, mobile applications have emerged as effective tools to support healthcare management. Smart solutions can assist users in organizing their medication routines, providing reminders, and maintaining records of their health activities. However, many existing systems are either too complex or not user- friendly, making them less effective for everyday use.

To address these challenges, the MEDLY application is designed as a simple, efficient, and user-friendly medication management system. It aims to bridge the gap between patients and proper medication adherence by offering features like reminders, tracking, and easy access to medication information.

This project focuses on developing a reliable platform that enhances user experience while ensuring better health outcomes through improved medication management.

1.1 Purpose

The primary purpose of the MEDLY application is to assist users in effectively managing their medication schedules and improving adherence to prescribed treatments. The application is designed to reduce the risk of missed doses, incorrect medication intake, and confusion related to multiple prescriptions.

MEDLY aims to serve as a digital companion for users by providing timely reminders, maintaining accurate records, and offering a structured approach to medication management. It focuses on simplifying complex medication routines into an easy-to-use system that can be accessed anytime and anywhere.Additionally, the purpose of this project is to demonstrate how technology can be integrated into healthcare to solve real- life problems. It highlights the importance of user-centered design and efficient system development in creating impactful digital solutions.

1.2 Problem Statement

Medication mismanagement is a widespread issue that affects millions of people worldwide. Many patients forget to take their medicines on time, take incorrect doses, or fail to follow prescribed schedules properly. This problem is especially common among elderly individuals and patients dealing with chronic diseases who need to manage multiple medications daily.

Lack of awareness, busy lifestyles, and absence of a proper tracking system contribute significantly to this issue. Traditional methods such as manual notes, alarms, or memory-based tracking are often unreliable and prone to errors..

Poor medication adherence can lead to serious consequences such as worsening health conditions,increased hospitalizations, and higher healthcare costs. It also reduces the effectiveness of treatments and can result in long-term complications. Therefore, there is a strong need for a simple, efficient, and accessible solution that helps users manage their medications accurately and consistently. The MEDLY application addresses this problem by providing an organized system for reminders, tracking, and monitoring medication usage.

1.3 Objectives

The main objective of the MEDLY application is to improve medication adherence and ensure better health outcomes through an easy-to-use digital platform. The application focuses on solving real- world problems by providing practical and efficient solutions.

The key objectives of MEDLY include:

- To provide a simple interface for users to add and manage their medications easily.
- To send timely reminders and notifications to ensure users do not miss their doses.
- To maintain a record of medication history for better tracking and monitoring.
- To reduce the chances of incorrect medication usage and improve safety.
- To enhance user experience through a clean, user-friendly design.
- To support individuals with busy schedules, elderly users, and patients with chronic conditions.

Through these objectives, MEDLY aims to create a reliable system that supports users in maintaining their
health effectively.

1.4 Scope

The scope of the MEDLY application includes the design and development of a medication management system that helps users organize and track their medication routines. The application focuses on core functionalities such as adding medications, setting reminders, and monitoring adherence.

The system is intended for a wide range of users, including patients, elderly individuals, and caregivers who need assistance in managing medication schedules. It can be used for both short-term and long-term treatments.

MEDLY is designed to be accessible on digital platforms, making it convenient for users to access their medication information anytime. The application aims to provide a seamless and efficient experience without requiring advanced technical knowledge.

However, the scope of the project is limited to medication tracking and management. It does not replace professional medical advice, diagnosis, or treatment. Future enhancements may include advanced features such as doctor integration, prescription uploads, and health analytics.

Additionally, MEDLY maintains a record of medication history, which can help users monitor their adherence over time. This feature can also be useful during medical consultations, as users can refer to their past medication patterns.

The scope also includes ensuring a smooth user experience through efficient system design, responsive interface, and reliable notification mechanisms. The application can be developed for mobile or web platforms, making it accessible anytime and anywhere.

Future scope of the application may include advanced features such as integration with healthcare providers, prescription uploads, AI-based health insights, and synchronization with wearable devices. These enhancements can further improve the effectiveness and usability of the system.

In summary, the scope of MEDLY is centered around creating a practical, scalable, and user-centric solution that simplifies medication management and contributes to better health practices.

Chapter 2: Proposed System

The system works by capturing medicine images and processing them to identify the medicine. After identification, the system displays medicine information and provides pharmacy location details. Users can also set reminders and track medicine stock to manage their medication effectively.

In addition to identification, the system provides a feature to locate nearby pharmacies. By using location-based services, MEDLY helps users find medical stores in their vicinity, making it easier to access medicines in case of emergencies or when stock runs out. This feature enhances convenience and ensures timely availability of required medications.

The application also includes a medication reminder system that allows users to set customized alerts based on their prescription schedules. Users can define the time, frequency, and dosage for each medicine, and the system will send timely notifications to ensure that no dose is missed. This feature is particularly beneficial for elderly users, patients with chronic diseases, and individuals managing multiple medications.

Another important component of the proposed system is medicine stock tracking. Users can monitor the availability of their medicines and receive alerts when stock levels are low. This prevents last-minute shortages and helps users maintain continuity in their treatment.

The proposed system ensures simple operation and easy accessibility for users.

2.1 Features and Functionality:

1. Structured Medicine Data Management

At its core, MEDLY uses a relational database (such as MySQL or PostgreSQL) to store all medicine- related data. Instead of dumping everything into flat files or unstructured storage, the data is split into well-defined tables with clear relationships between them.

For example:

- A medicines table stores medicine ID, name, manufacturer, category, and description
- A dosages table links to the medicine and holds strength (e.g., 500mg), form (tablet/syrup), and frequency A medicine_details table might hold side effects, contraindications, and storage instructions

Each table has primary keys (unique identifiers) and foreign keys (links to related tables), enforcing referential integrity. This means you can never have a dosage record that points to a non-existent medicine. Queries like "give me all dosages for Paracetamol"

2. Intelligent Medicine Search

A naive search would only find exact matches — useless if a user types "Paracetamol" as "Parcetamol". MEDLY solves this with two layers:

String Matching techniques like substring search (LIKE 'para' in SQL) or prefix matching catch partial inputs. If a user types "amox", it surfaces "Amoxicillin", "Amoxiclav", etc.

Fuzzy Search algorithms like Levenshtein Distance (edit distance) measure how many character insertions, deletions, or substitutions are needed to transform one word into another. "Ibuprofn" has an edit distance of 1 from "Ibuprofen", so it still matches. Libraries like FuzzyWuzzy (Python) or ElasticSearch implement this at scale. The system ranks results by similarity score and returns the top candidates, making the search forgiving of typos and phonetic variations.

3. Efficient Dosage Reminder Scheduling

When a user sets up a reminder, they provide:

- Medicine name (linked to the DB) Dosage (how many tablets/ml)
- Frequency (once daily, twice daily, every 8 hours, etc.)
- Duration (for 7 days, 30 days, ongoi\ng)
- Start time (e.g., 8:00 AM)

The system stores these as scheduled jobs in a reminders table with timestamps. A background scheduler (like APScheduler in Python or cron jobs) continuously checks this table and triggers push notifications or in-app alerts at the right times. The scheduling logic also handles edge cases — like what happens if the user installs the app mid-course of a prescription, or if they pause reminders for a day.

4. Missed Dose Tracking and Alerts

Every scheduled reminder has a corresponding acknowledgment mechanism — the user marks a dose as "taken." If no acknowledgment arrives within a set window (say, 30 minutes after the reminder), the system flags that dose as missed and logs it in a dose_logs table with a status field (taken / missed / skipped). Over time, this builds an adherence profile. If a user misses more than 2 consecutive doses, the system escalates — sending a stronger alert or a follow-up reminder. Aggregate statistics (e.g., "You missed 4 of 14 doses this week — 71% adherence") can be shown on a dashboard. This is clinically significant: poor adherence is a leading cause of treatment failure.

5. Drug Interaction Detection

This is a safety-critical feature. Certain drug combinations are dangerous — for example, taking blood thinners (Warfarin) with Aspirin increases bleeding risk. MEDLY maintains a drug interaction table with pairs of medicines and their known interaction severity (mild / moderate / severe).

When a user adds a new medicine to their schedule, the system queries this table against all medicines the user is currently taking. If a match is found, it immediately raises a warning with a plain-language explanation (e.g., "Taking Metformin with Alcohol may cause lactic acidosis"). The rules are predefined by medical guidelines (like those from the FDA or WHO drug interaction databases). This rule-based system is not AI — it's deterministic lookup, which makes it predictable and auditable.

6. Symptom-Based Medicine Recommendation

This feature uses a rule-based inference engine — essentially a structured if-then knowledge base built from medical guidelines. The user selects symptoms (e.g., headache + fever + body ache), and the system runs these through a set of rules: IF symptom = fever AND symptom = body_ache THEN suggest = Paracetamol

IF symptom = congestion AND symptom = runny_nose THEN suggest = Cetirizine

These rules are stored in a symptom_medicine_map table or a rules file. The engine finds all rules that match the selected symptom set and returns the union of suggested medicines, ranked by how many symptoms each one addresses. This is basic forward chaining logic — not a neural network, but transparent, explainable, and easy to audit.

7. Disease Mapping and Treatment Suggestions

This goes a step deeper than symptom-medicine mapping. Here, the system maps symptom combinations to diseases, then maps diseases to treatment protocols. The scoring logic works like this:

Each disease has a predefined symptom profile (e.g., Malaria: fever, chills, sweating, headache)

When a user reports symptoms, the system calculates an overlap score for each disease (how many of that disease's symptoms the user reported)

Diseases are ranked by score; the top candidates are presented as likely conditions For example, if a user reports fever + chills + sweating, Malaria might score 3/4 while Common Cold scores 1/4. The system then suggests treatments linked to the top-scoring disease. This is essentially a weighted symptom matching algorithm

— simple but effective for a first-pass triage tool.

8. Pharmacy Data Management

Rather than calling a live external API (like Google Places) every time, MEDLY maintains its own pharmacy database internally. Each record contains:

GPS coordinates (latitude & longitude) Available medicine categories

This data is pre-populated and periodically updated. The advantage is zero dependency on third-part services — the app works offline or in low-connectivity environments. The tradeoff is that the data must be maintained manually or via periodic sync jobs. For a localized deployment (e.g., a specific city or hospital network), this is very practical.

9. Nearest Pharmacy Locator

Once the user's GPS coordinates are known, MEDLY uses the Haversine Formula to compute the great-circle distance between the user and every pharmacy in the database.

The formula accounts for Earth's curvature:

$a = \sin^2(\square lat/2) + \cos(lat1) \times \cos(lat2) \times \sin^2(\square lon/2)$ distance $= 2R \times \arcsin(\square a)$ Where R is Earth's radius (~6,371 km).

This is computed for every pharmacy record, and the results are sorted ascending by distance. The top 3–5 nearest pharmacies are displayed to the user with their distance and contact info. This is done entirely
in-app — no Google Maps API needed — making it cost-free and privacy-friendly.

10. User History and Records Management

Every meaningful action a user takes is logged in a user_history table:

Which medicines they took and when

Which reminders they acknowledged or

missed What symptoms they reported

What searches they performed

What schedules they created or modified

This creates a longitudinal health timeline for each user. The data can be visualized as charts (weekly adherence rate, dose history calendar) or exported as a PDF report for a doctor's visit. It also enables the system to personalize recommendations over time — for instance, noticing that a user consistently misses their 2 PM dose and suggesting they shift it to 3 PM.

11. Data Consistency and Integrity

MEDLY's database is normalized — typically to at least Third Normal Form (3NF). This

means: No repeating groups — medicine details aren't copy-pasted across multiple rows

No partial dependencies — every non-key column depends on the whole primary key

No transitive dependencies — columns only depend on the primary key, not on other non-key columns

In practice, this means if a medicine's name changes, you update it in one place (the medicines table) and every related record (dosages, reminders, logs) automatically reflects that change. Constraints like NOT NULL, UNIQUE, FOREIGN KEY, and CHECK are enforced at the database level — not just in application code — ensuring that bad data can never enter the system even if there's a bug in the app logic.

Together, these 11 features make MEDLY a comprehensive, safety-conscious, and technically sound medication management system built on solid software engineering and database design principles

Chapter 3: Project Outcomes

This chapter presents the key results and deliverables achieved throughout the project lifecycle. It evaluates the extent to which objectives were met, highlighting both successes and areas for improvement.

Improved Medication Adherence

The Medly application plays a crucial role in improving medication adherence by ensuring that users strictly follow their prescribed schedules. Through automated and customizable reminders, users are notified at the exact time they need to take their medication. This reduces dependency on memory and minimizes human error. Additionally, the system tracks whether doses are taken or missed, helping users stay accountable. This is particularly beneficial for patients with chronic conditions such as diabetes or hypertension, where consistent medication intake is critical for effective treatment and prevention of complications.

Better Health Management

Medly enables users to manage their health in a more structured and systematic manner. By consolidating all medication-related tasks—such as scheduling, tracking, and reviewing—into a single platform, users can maintain a disciplined healthcare routine. The application allows continuous monitoring of medication habits, which helps in identifying irregularities and improving consistency. Over time, this leads to better treatment outcomes, reduced health risks, and an overall improvement in quality of life.

Increased User Awareness

One of the key outcomes of the system is the enhancement of user awareness regarding medications. Medly provides detailed and easily understandable information about each medicine, including its purpose, dosage instructions, precautions, and potential side effects. This helps users make informed decisions and reduces confusion related to medication usage. Increased awareness also encourages responsible behavior, ensuring that users do not misuse or overuse medicines, thereby promoting safer healthcare practices.

Reduced Risk of Medication Errors

Medication errors can lead to serious health complications. Medly addresses this issue by integrating multiple safety features such as dosage tracking, reminder notifications, and drug interaction alerts. The system ensures that users take the correct medicine at the correct time and in the correct quantity. It also warns users about possible harmful interactions between different medicines. By providing these safeguards, the application significantly reduces the chances of errors like overdosing, missed doses, or incorrect combinations, thereby enhancing patient safety.

Enhanced Convenience and Accessibility

Medly offers a highly convenient solution by bringing all medication-related functionalities into a single, easy-to-use platform. Users no longer need to rely on multiple applications, physical notes, or memory to manage their medications. All essential information—such as prescriptions, schedules, reminders, and pharmacy details—is readily accessible anytime and anywhere. This improves user experience and ensures that healthcare management becomes seamless, efficient, and stress-free.

Efficient Pharmacy Access

The application includes a pharmacy locator feature that allows users to find nearby medical stores quickly and efficiently. Using location-based calculations, the system identifies the closest pharmacies and provides relevant details. This is particularly useful during emergencies or when medicines are urgently required. By reducing the time spent searching for pharmacies, Medly ensures that users can access medications promptly, which can be critical in time-sensitive situations.

Data-Driven Health Insights

Medly continuously records user activity, including medication schedules, adherence patterns, and history. This data is valuable for generating insights into user behavior. By analyzing these patterns, users can identify trends such as missed doses or irregular timings. These insights enable users to improve their habits and make informed health decisions. Over time, such data-driven analysis contributes to better long-term health management and preventive care.

Time-Saving and Organized System

The automation provided by Medly significantly reduces the time and effort required to manage medications. Users are not required to manually track schedules or maintain records, as the system handles everything automatically. This not only saves time but also reduces cognitive load, especially for individuals managing multiple medications. The organized structure of the application ensures that all information is systematically stored and easily retrievable, improving overall efficiency.

Support for Diverse User Groups

Medly is designed to be inclusive and user-friendly, catering to a wide range of users with different needs. Elderly users benefit from reminders and simplified interfaces, reducing the risk of forgetting medications. Busy professionals can rely on automation to manage their schedules without disruption. Patients with chronic illnesses benefit from consistent tracking and monitoring. This

adaptability makes the application suitable for diverse demographics, enhancing its usability and effectiveness.

Reliable and Independent Functionality

A major strength of Medly is its ability to function independently using internally implemented algorithms. Features such as fuzzy search for medicine names, rule-based systems for recommendations, and distance calculations for pharmacy location are all handled within the system. This reduces dependency on external APIs, ensuring faster performance, improved reliability, and consistent functionality even in low or no internet conditions. This makes the system robust, efficient, and suitable for real-world usage.

Chapter 4: Software Requirements

The MEDLY application requires a combination of modern frontend, backend, database, and API technologies to ensure smooth performance, scalability, and a user-friendly experience. The following are the detailed software requirements:

Frontend Requirements

The frontend of MEDLY is developed using Flutter 3.x, which provides a cross-platform UI framework for building high-performance applications for both Android and iOS. The programming language used is Dart (null-safe), ensuring type safety and reducing runtime errors.

The application follows Material Design 3 principles to deliver a clean, modern, and intuitive user interface. It ensures consistency in design, better user experience, and easy navigation.

Additional Flutter packages used include:

flutter_local_notifications – to implement scheduled reminders and push notifications for medication alerts.

url_launcher – to enable external actions such as opening maps for pharmacy locations or initiating calls. flutter/services – to handle device-level functionalities such as clipboard access and system interactions.

Backend Requirements

The backend logic is implemented using Dart, utilizing asynchronous programming concepts like async/await to ensure smooth and non-blocking operations.

1. The application follows a structured architecture including:

- Service Layer Pattern – to separate business logic from UI.
- Singleton Pattern – to ensure a single instance of key services like database handlers.
- Repository Pattern – to manage data access and improve maintainability. Core algorithms implemented in the backend include:
- Levenshtein Distance Algorithm – for fuzzy search, allowing users to find medicines even with incorrect spellings.
- Haversine Formula – to calculate the distance between the user and nearby pharmacies accurately.
- Weighted Symptom Scoring – a rule-based system to map symptoms to possible conditions and suggest treatments.
- Pair-wise SQL Lookup – used to detect drug interactions between multiple medicines.

Additional packages:

- http – for making API calls to external services.
- The database is designed using Third Normal Form (3NF) to maintain data integrity, reduce redundancy, and improve performance.

2. Key database features include:

- Use of foreign keys and JOIN operations for relational data

  handling. Implementation of the LIKE operator for efficient text-based

  search queries.

- Use of path_provider and path packages to manage database file storage within the

  device. The application maintains multiple local databases for modular data handling:

- Medicines database

  Reminders database

- Drug interactions database
- Conditions (symptoms & diseases) database Prescriptions database

API and Platform Requirements

The system integrates with the Geoapify Places API to provide pharmacy search functionality. This API returns data in GeoJSON format, including details such as pharmacy name, location coordinates, and contact information.

MEDLY is designed as a cross-platform application, supporting both:

Android devices

iOS devices

This ensures wider accessibility and usability across different user groups.

Backend Requirements

The backend logic is implemented using Dart, utilizing asynchronous programming concepts like

async/await to ensure smooth and non-blocking operations. The application

follows a structured architecture including:

- Service Layer Pattern – to separate business logic from UI.
- Singleton Pattern – to ensure a single instance of key services like database handlers.
- Repository Pattern – to manage data access and improve maintainability. Core algorithms implemented in the backend include:
- Levenshtein Distance Algorithm – for fuzzy search, allowing users to find medicines even

with incorrect spellings.

- Haversine Formula – to calculate the distance be$_{1}$t$_{8}$ween the user and nearby pharmacies accurately.
- Weighted Symptom Scoring – a rule-based system to map symptoms to possible conditions and suggest treatments.
- Pair-wise SQL Lookup – used to detect drug interactions between multiple medicines.
  Additional packages:
- http – for making API calls to external services.
- geolocator – for accessing GPS location and handling permissions.

Database Requirements

MEDLY uses SQLite (sqflite) as its local database system, ensuring fast, offline, and efficient data storage.

The database is designed using Third Normal Form (3NF) to maintain data integrity, reduce redundancy, and improve performance.

Key database features include:

- Use of foreign keys and JOIN operations for relational data handling. Implementation of the LIKE operator for efficient text-based search queries.
- Use of path_provider and path packages to manage database file storage within the device. Medicines database
- Reminders database
- Drug interactions database
- Conditions (symptoms & diseases) tabase Prescriptions database

API and Platform Requirements

The system integrates with the Geoapify Places API to provide pharmacy search functionality. This API returns data in GeoJSON format, including details such as pharmacy name, location coordinates, and contact information.

MEDLY is designed as a cross-platform application, supporting both:

Android devices

iOS devices

This ensures wider accessibility and usability across different user groups.

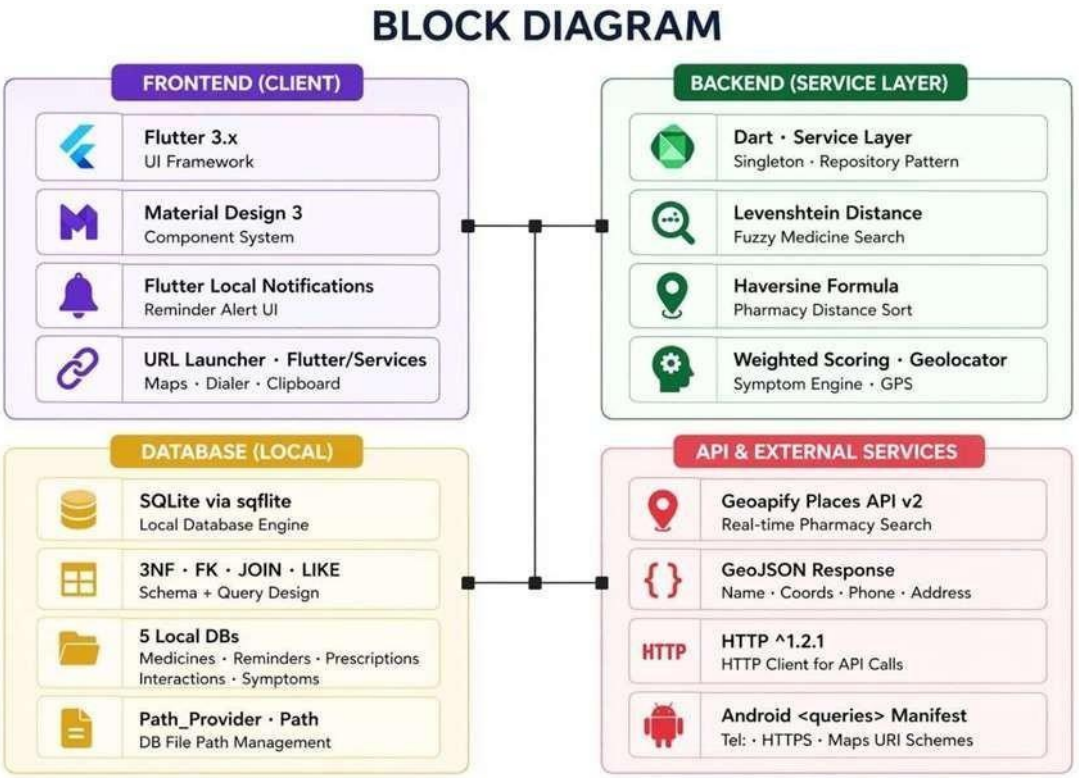

Figure 1 – Block representation of Required Softwares

Figure 1 represents the overall architecture of the Medly application, divided into four main components: Frontend, Backend, Database, and API services. The Frontend handles user interaction using Flutter, while the Backend processes logic using algorithms like fuzzy search and distance calculation.

The Database stores all application data locally using SQLite. The API layer enables external communication for features like pharmacy location. Together, these components ensure smooth data flow and efficient functioning of the system.

Chapter 5: Project Design

The design of the MEDLY application focuses on creating a scalable, efficient, and user-friendly system that integrates medicine management, intelligent search, and location-based services. The project follows a modular and layered architecture to ensure maintainability, flexibility, and performance.

1. System Architecture

MEDLY is designed using a layered architecture that separates different functionalities into distinct
layers. This improves code organization and makes the system easier to manage and extend.

• Presentation Layer (UI)

Built using Flutter, this layer handles user interaction, input, and display of information. It ensures smooth and responsive user experience using Material Design principles.

• Business Logic Layer (Service Layer)

This layer processes user inputs, applies rules, and executes core functionalities such as reminders, symptom analysis, and medicine identification.

• Data Access Layer (Repository Layer)

Responsible for managing communication between the application and the database. It fetches, inserts,
updates, and deletes data efficiently.

• Database Layer

Uses SQLite to store structured data such as medicines, reminders, and user history.

2. Modular Design Approach

The system is divided into independent modules, each responsible for a specific functionality. This modular approach enhances reusability and simplifies debugging.

• Medicine Management Module – Handles storage, retrieval, and display of medicine information.

• Search Module – Implements fuzzy search using the Levenshtein distance algorithm.

• Reminder Module – Manages scheduling and triggering of medication alerts.

• Symptom Analysis Module – Uses rule-based logic to map symptoms to diseases.

• Pharmacy Locator Module – Finds nearby pharmacies using GPS and distance calculations.

• Interaction Checker Module – Detects potential drug interactions using database rules.

3. Data Flow Design

The data flow in MEDLY ensures smooth communication between components:

1. User inputs data (medicine details, symptoms, etc.) through the UI.
2. The input is sent to the service layer for processing.
3. The service layer applies algorithms and business rules.
4. Required data is fetched or stored in the SQLite database via the repository layer.
5. Results are returned to the UI and displayed to the user.

This structured flow ensures efficient processing and reduces system complexity

4. Database Design

The database is designed using relational modeling and normalization (3NF) to ensure data consistency and reduce redundancy.

• Multiple tables are used for different entities such as medicines, reminders, interactions, and conditions.

• Relationships are maintained using foreign keys.

• Efficient queries are implemented using JOIN operations and search using the LIKE operator. This design ensures fast data retrieval and reliable storage.

• MEDLY integrates several algorithms to enhance functionality:

• Levenshtein Distance – Enables fuzzy search for medicine names.

• Haversine Formula – Calculates distance between user and pharmacies.

• Weighted Scoring Algorithm – Maps symptoms to possible diseases.

• Pair-wise Comparison Logic – Detects drug interactions.

These algorithms improve accuracy, usability, and intelligence of the system.

5. User Interface Design

The UI is designed with a focus on simplicity and accessibility:

• Clean layout using Material Design 3

• Easy navigation with minimal steps

• Clear icons, buttons, and readable text

• Responsive design for different screen sizes

This ensures that users of all age groups can easily interact with the application.

6. Security and Reliability Design

• Data is stored locally using SQLite to ensure privacy.

• Input validation is implemented to avoid incorrect data entry.

• Error handling mechanisms are used to maintain system tability.

7. Scalability and Future Enhancements

The design allows future improvements such as:

• Integration with cloud databases

• AI-based recommendations

• Doctor and prescription integration

Chapter 6: Project Scheduling

The development of the MEDLY application was carried out in a structured and systematic manner over a period of four months. Each phase of the project focused on specific activities to ensure proper planning, development, and successful implementation.

Project Timeline & Progress (Month-wise + Week-wise Explanation)

JANUARY – Problem Identification and Objective Definition (Week 1–2)

During the initial phase of the project, the focus was on identifying the real-world problem of medication mismanagement. Research was conducted to understand issues such as missed doses, lack of proper tracking, and difficulty in managing multiple medicines.

In Week 1, the project team was formed and the topic ―Medly‖ was finalized. The scope and objectives of the system were defined clearly. In Week 2, requirement analysis was performed where key features like medicine search, reminders, symptom checker, and pharmacy locator were identified.

Based on this analysis, the objectives of the application were finalized, and initial documentation along with requirement gathering was completed. This phase laid the foundation for the entire project.

FEBRUARY – System Design and Planning (Week 3–5)

In this phase, the system architecture and design were developed using a layered approach to ensure scalability and maintainability.

In Week 3, the project idea and functionalities were discussed with the guide, and necessary improvements were made. In Week 4, the graphical user interface (GUI) was designed, including layouts and navigation flow using Flutter principles. In Week 5, the first presentation was conducted to demonstrate the proposed system design and planning.

Detailed planning included:

Database design (tables, relationships, normalization)

Module structure (reminders, interaction checker, pharmacy locator, etc.)

Algorithm selection (fuzzy search, Haversine formula, rule-based logic) This phase ensured that development would be structured and efficient.

MARCH – Development of Core Modules (Week 6–10)

This phase involved the actual implementation and development of the Medly application.

In Week 6, the database was designed using SQLite with proper normalization techniques. In Week 7, backend logic and core algorithms such as fuzzy search and rule-based systems were implemented.

In Week 8, initial integration of modules like medicine service, reminder system, and symptom checker was carried out. In Week 9, database connectivity was established, enabling data storage and retrieval across all modules.

In Week 10, advanced features such as pharmacy locator (using Haversine formula), interaction checker, and adherence tracking were implemented.

Key development activities included:

Implementation of search and interaction checking logic
Development of reminder system using local notifications

Creation of database with CRUD operations

Integration of all major modules

Each module was tested individually to ensure correct functionality before integration.

APRIL – Testing and Report Preparation (Week 11–12)

The final phase focused on testing, debugging, and documentation of the project. In Week 11, all modules were fully integrated, and testing was performed to identify and fix errors. UI improvements were also made based on feedback

In Week 12, the final presentation was conducted, and the complete project report was prepared and submitted.

Key activities included:

Debugging and fixing issues in all modules

Verifying accuracy of reminders, search, and pharmacy locator

Enhancing user interface and performance

Preparing final documentation and report

Creating PPT and demonstrating the project

This phase ensured that the Medly application was reliable, efficient, and ready for evaluation.

The project was completed in a systematic and structured manner, progressing from problem identification to design, development, and final testing. The combination of monthly planning and weekly execution ensured timely completion, proper coordination, and successful implementation of the Medly application.

Figure 3 : Gantt chart of medly

Figure 3 represents a Gantt Chart for the MEDLY project, showing the timeline and progress of different tasks.It divides the project into phases like Project Conception and Design & Implementation, with specific activities listed under each.Each colored bar indicates the duration and schedule of tasks across weeks.It also shows task owners, start/end dates, and completion percentage.Overall, it helps track project planning, progress, and dependencies visuals

Chapter 7: Results

The MEDLY application was successfully developed and tested, demonstrating effective performance across all its core functionalities. The system provides a reliable and user- friendly platform for medication management and healthcare support.

The medicine management module allows users to add, update, and view medicine details efficiently. Data is stored securely in the SQLite database and retrieved quickly, ensuring smooth performance without delays.

The intelligent search feature using fuzzy search (Levenshtein distance) works accurately, enabling users to find medicines even with spelling errors or incomplete input. This improves usability and reduces dependency on exact search terms.

The reminder system functions effectively by sending timely notifications based on user- defined schedules. It ensures that users receive alerts for each dose, helping in reducing missed medications and improving adherence.

The medicine scanner functionality successfully identifies medicines based on input and displays relevant details. This enhances user awareness and simplifies medicine identification.

The pharmacy locator module accurately identifies nearby pharmacies using GPS and the Haversine formula. The integration with location services provides real-time and relevant results, improving accessibility to medicines.

The drug interaction checker successfully detects potential interactions between selected medicines and alerts users, contributing to safer medication usage.

The symptom-based recommendation system provides basic suggestions by mapping symptoms to possible conditions using rule-based logic. This feature assists users in making informed decisions.

The system maintains user history and adherence records, allowing users to track their medication patterns over time. This helps in better health monitoring and future reference.

Overall, the application performs efficiently without reliance on heavy external APIs, ensuring reliability, faster response time, and offline capabilities for core features.

Figure 4: Medicine Search Interface of Medly Application

Figure 4 shows the medicine search feature of the Medly application. The user has entered the keyword—aspi‖ in the search bar, and the system dynamically displays relevant results such as Aspirin. Each result is presented in a clean card layout with an icon and navigation arrow, allowing users to view detailed information. The interface also includes a bottom navigation bar with options like Search, Symptoms, Remind, Interact, Pharmacy, and Profile for easy access to other features. This screen demonstrates the app's user-friendly design and real-time search functionality for quick medicine lookup.

Figure 5: All Medicines Listing Screen of Medly Application

figure 5 displays the ―All Medicines‖ section of the Medly app, where users can browse a list of commonly used medicines such as Paracetamol, Ibuprofen, Amoxicillin, Cetirizine, and Omeprazole.Each medicine card shows important details like dosage, type (tablet/capsule), and its medical category.The search bar at the top allows users to quickly find specific medicines. This screen helps users explore, identify, and select medicines easily, improving accessibility and user experience within the application.

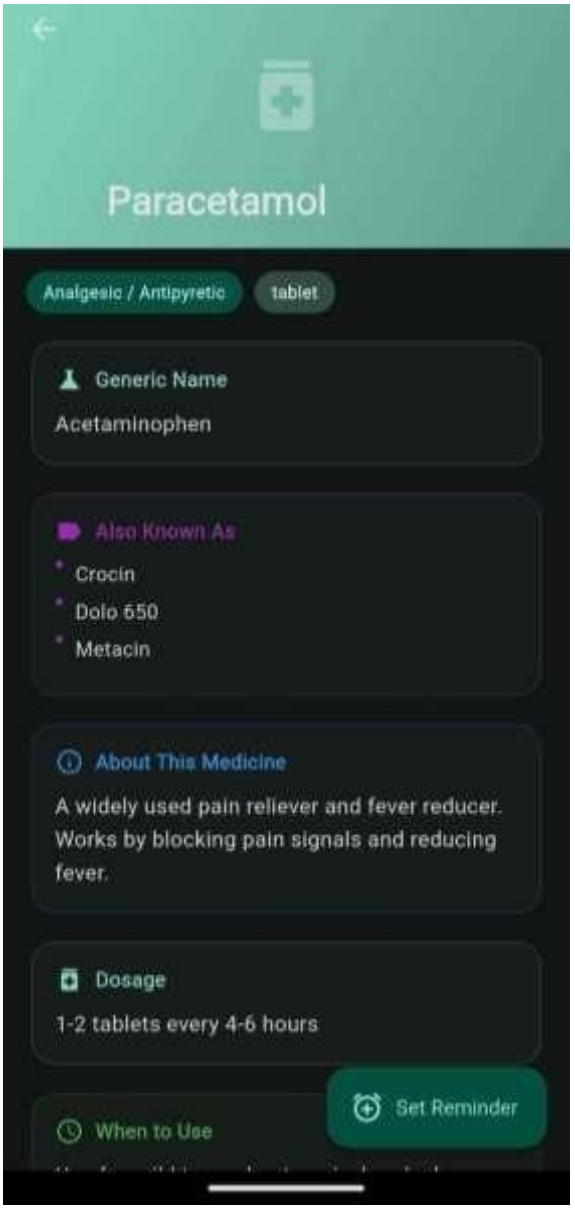

Figure 6: Medicine Detail Screen of Medly Application

figure 6 displays the detailed information screen for Paracetamol within the Medly application. It presents comprehensive medicine data including its generic name (Acetaminophen), popular brand names it is also known as (Crocin, Dolo 650, Metacin), a brief description of its use, recommended dosage (1-2 tablets every 4-6 hours), and its category as an Analgesic/Antipyretic tablet. The screen is designed to give users a complete overview of a selected medicine at a glance, supporting informed and safe medication usage

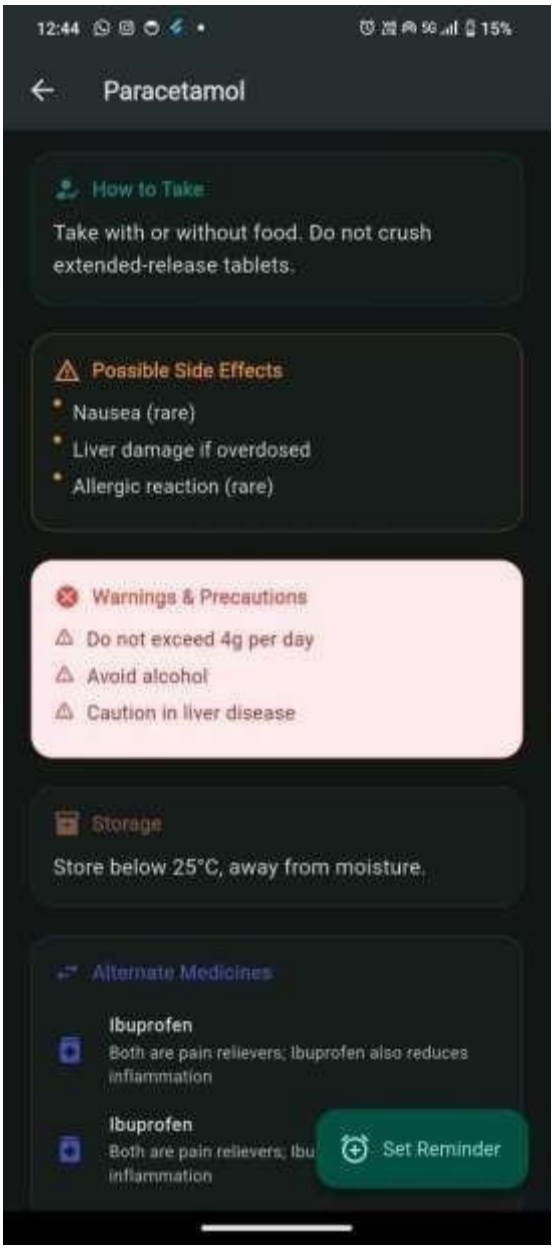

Figure 7: Medicine Safety & Storage Detail Screen of Medly Application

figure 7 shows the extended detail view of Paracetamol in the Medly application, displaying essential safety-related information for the user. It includes sections for How to Take, Possible Side Effects, and a prominently highlighted Warnings & Precautions block in red, alerting users about critical limits such as not exceeding 4g per day and avoiding alcohol. The Storage section advises users to keep the medicine below 25°C away from moisture, ensuring proper handling. At the bottom, an Alternate Medicines section suggests similar drugs like Ibuprofen along with a brief comparison, helping users make informed choices when the primary medicine is unavailable.

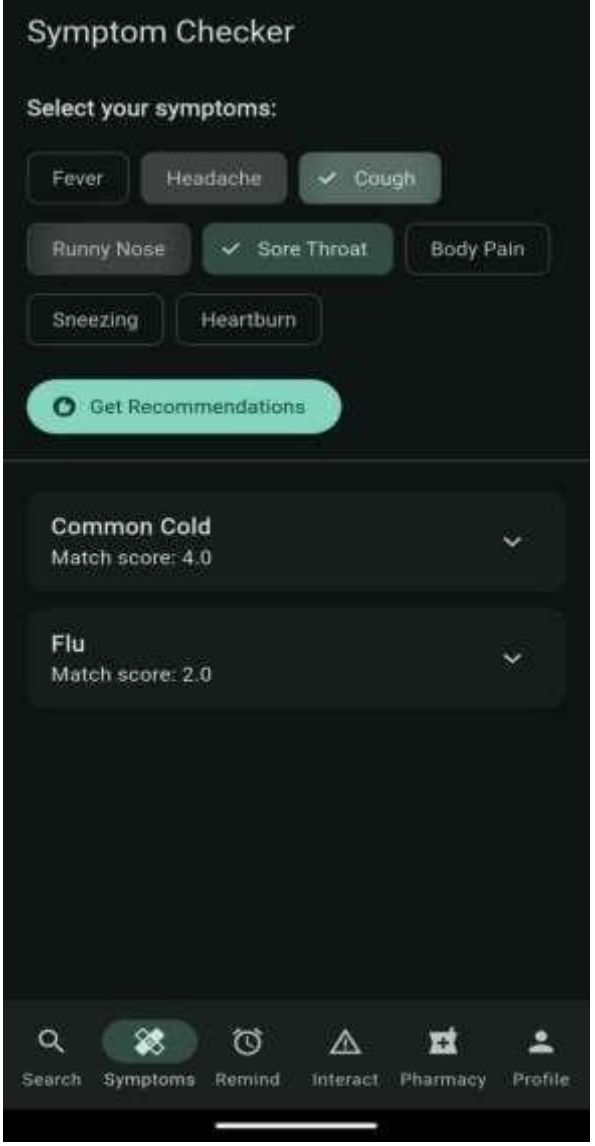

Figure 8: Symptom Checker Screen of Medly Application

figure 8 displays the Symptom Checker screen of Medly, where users can select their current symptoms such as Cough and Sore Throat from a predefined list and tap "Get Recommendations" to receive results. The system uses a rule-based scoring algorithm to match the selected symptoms with possible diseases, displaying Common Cold (score: 4.0) and Flu (score: 2.0) as the most likely conditions. This feature provides users with a quick and intelligent preliminary health assessment, helping them identify possible illnesses and take informed medication decisions.

Figure 9: Medicine Reminders Screen of Medly Application

figure 9 showcases the Medicine Reminders screen of Medly, which displays all scheduled medication reminders with their timings, medicine names, frequency, and current status — marked as either "Taken" (green) or "Missed" (red), as seen for Paracetamol and Amoxicillin throughout the day. Users can manually update each dose status using the Taken or Missed buttons, and add new reminders via the "Add Reminder" button, enabling effective tracking of daily medication adherence.

Figure 10: Drug Interaction Checker Screen of Medly Application

Figure 10 displays the Drug Interaction Checker screen of Medly, where users can select multiple medicines (here, 4 selected including Ibuprofen and Aspirin) and tap "Check Interactions" to analyze potential conflicts between them. The results are displayed with clear safety labels — combinations like Ketorolac + Aspirin and Doxycycline + Ibuprofen are marked "SAFE", while Ibuprofen + Aspirin is flagged as "MEDIUM" risk, warning that Ibuprofen can block the antiplatelet effect of low-dose Aspirin when taken simultaneously, along with a suggested precaution. This feature acts as a critical safety tool within Medly, helping users avoid harmful drug combinations and make safer medication decis

Figure 11: My Health Profile Screen of Medly Application

figure 11 displays the My Health Profile screen of Medly, where users can store and manage their personal health information including name, age, gender, weight, height, and blood group. It also includes a Medical Conditions section where users can indicate existing conditions such as Blood Pressure (here set to Low BP) and Diabetes status, helping the app personalize health recommendations. A Low Stock Alert at the top notifies the user that Amoxicillin has only 1 tablet remaining, prompting a timely refill, while the "Track Medicine" button allows users to monitor their ongoing medications directly from the profile screen.

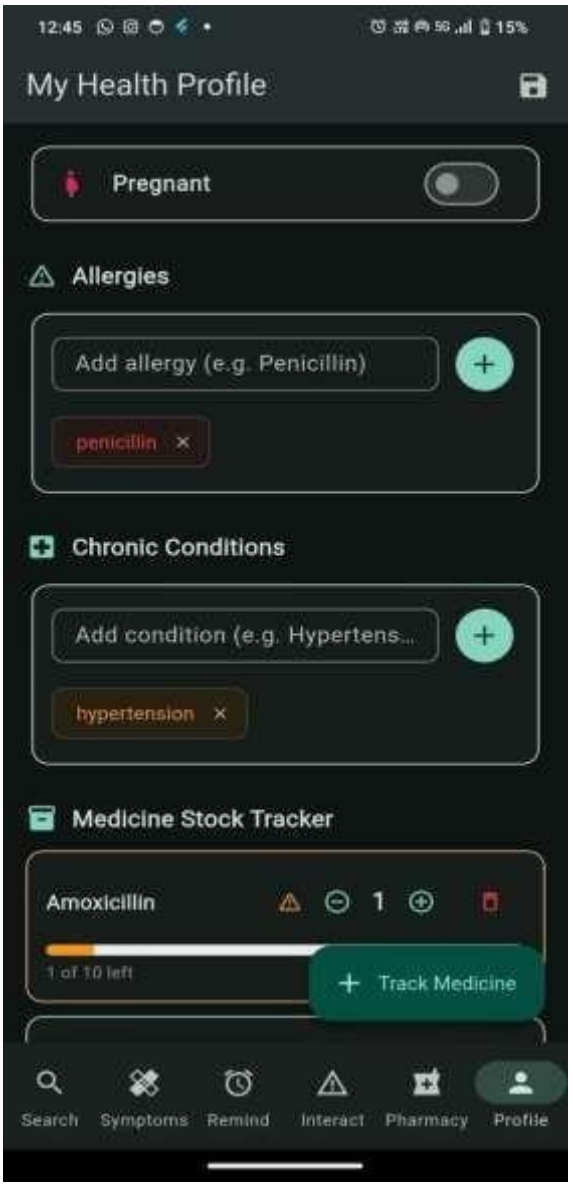

Figure 12: Health Profile Extended & Medicine Stock Tracker Screen of Medly Application

figure 12 shows the extended section of the My Health Profile screen, where users can record additional health details such as Pregnancy status, known Allergies (here, Penicillin is added), and Chronic Conditions (here, Hypertension is listed). At the bottom, the Medicine Stock Tracker displays the remaining quantity of each medicine — Amoxicillin shows only 1 of 10 tablets left with a low-stock warning, visually represented by a progress bar. This feature ensures users are always aware of their medicine inventory and can refill on time, preventing missed doses due to stock \shortage

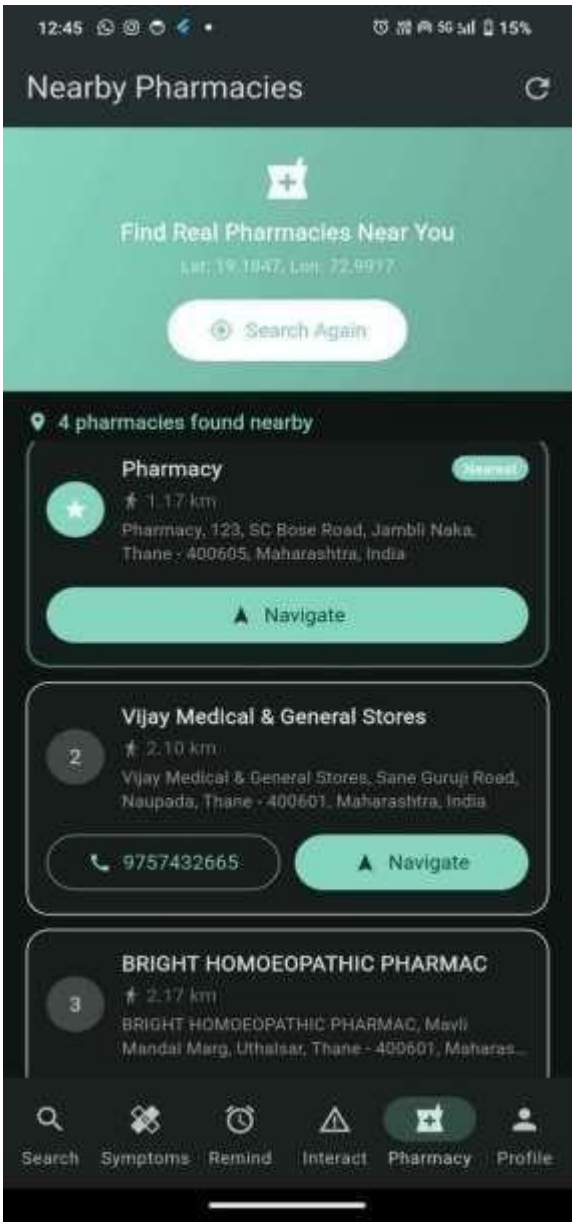

Figure 13: Nearby Pharmacies Screen of Medly Application

figure 13 displays the Nearby Pharmacies screen of Medly, which uses the user's current GPS coordinates (Lat: 19.1847, Lon: 72.9917) and the Haversine formula to calculate and list the closest pharmacies ranked by distance. A total of 4 pharmacies are found nearby, with the nearest one at just 1.17 km, followed by Vijay Medical & General Stores at 2.10 km and Bright Homoeopathic Pharmacy at 2.17 km, each displaying their full address, contact number, and a Navigate button for directions. This feature helps users quickly locate and reach the nearest medical

Chapter 8: Conclusion

The MEDLY – Medicine Scanner Application provides a simple and effective solution for medicine identification and medication management by focusing on four main objectives: medicine image scanning, detailed medicine information, nearby pharmacy location, and medication reminder with tracking. The system helps users easily scan medicines using a camera or gallery image and quickly identify them, reducing confusion and saving time.

The application successfully provides important medicine details such as usage, dosage, side effects, and alternative medicines, which helps users understand medicines properly and use them safely. The pharmacy locator feature allows users to find nearby medical stores and access medicines easily when required, improving convenience and accessibility.

The medication reminder and tracking feature ensures that users take medicines on time and maintain proper medication schedules by providing notifications and stock tracking. This helps users manage their medicines efficiently and avoid missing doses.

Overall, the MEDLY Medicine Scanner Application successfully achieves its objectives by providing a user-friendly and reliable system for medicine identification and medication management. The project demonstrates an effective digital solution that supports users in scanning medicines, viewing information, locating pharmacies, and managing medication in a simple and structured manner

References :

- Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. International Conference on Machine Learning (ICML). (Used for medicine image classification backbone)
- Dayer, L., et al. (2013). Smartphone Medication Adherence Apps: Potential Benefits to Patients and Providers. Journal of the American Pharmacists Association, 53(2), 172–181. (Supports the reminder and medication tracking module)
- Jin, R., et al. (2020). Pill Recognition Using Imaged Deep Learning for Smart Medication Management. Sensors, 20(9), 2455. MDPI. (Directly related pill/medicine recognition using deep learning on mobile)